6.5 Work-Energy Principle in curvilinear motion

Consider the equations of curvilinear translation

$$\frac{W}{g}\ddot{x} = \Sigma F_x$$

$$\frac{W}{g}\dot{x}\frac{d\dot{x}}{dx} = \Sigma F_x$$

$$\frac{W}{g}\ \dot{x}\, d\dot{x} = \Sigma F_x\, dx$$

Integrating both sides knowing that the body starts from origin with a x-velocity component $\dot{x}_0$

$$\frac{W}{g}\int_{\dot{x}_0}^{\dot{x}} \dot{x}\, d\dot{x} = \int_0^x \Sigma F_x\,.dx$$

$$\frac{W}{g}\frac{\dot{x}^2}{2} - \frac{W}{g}\frac{{\dot{x}_0}^2}{2} = \int_0^x \Sigma F_x\,.dx$$

$$\frac{W}{g}\ddot{y} = \Sigma F_y$$

$$\frac{W}{g}\dot{y}\frac{d\dot{y}}{dy} = \Sigma F_y$$

$$\frac{W}{g}\ \dot{y}\, d\dot{y} = \Sigma F_y\, dy$$

Integrating both sides knowing that the body starts from origin with a y-velocity component $\dot{y}_0$

$$\frac{W}{g}\int_{\dot{y}_0}^{\dot{y}} \dot{y}\, d\dot{y} = \int_0^y \Sigma F_y\,.dy$$

$$\frac{W}{g}\frac{\dot{y}^2}{2} - \frac{W}{g}\frac{{\dot{y}_0}^2}{2} = \int_0^y \Sigma F_y\,.dy$$

Adding these two equations

$$\frac{W}{g}\frac{\dot{x}^2}{2} - \frac{W}{g}\frac{{\dot{x}_0}^2}{2} + \frac{W}{g}\frac{\dot{y}^2}{2} - \frac{W}{g}\frac{{\dot{y}_0}^2}{2} = \int_0^x \Sigma F_x\,.dx + \int_0^y \Sigma F_y\,.dy$$

$$\frac{W}{g}\frac{(\dot{x}^2 + \dot{y}^2)}{2} - \frac{W}{g}\frac{({\dot{x}_0}^2 + {\dot{y}_0}^2)}{2} = \int_0^x \Sigma F_x\,.dx + \int_0^y \Sigma F_y\,.dy$$

$$\frac{W}{g}\frac{v^2}{2} - \frac{W}{g}\frac{{v_0}^2}{2} = \int_0^x \Sigma F_x\,.dx + \int_0^y \Sigma F_y\,.dy$$

Change in kinetic energy of the body= Net work done by the forces.

We can also write the equation of motion in the form

$$\frac{W}{g}a_n = \Sigma F_n$$

ΣF_n is the net force in the normal direction to the path. The work done by the normal force is zero always.

$$\frac{W}{g}a_t = \Sigma F_t$$

ΣF_t is the net force in the tangential direction to the path. $a_t = \frac{dv}{dt} = \frac{dv}{ds}\frac{ds}{dt} = v\frac{dv}{ds}$

$$\frac{W}{g}v\frac{dv}{ds} = \Sigma F_t$$

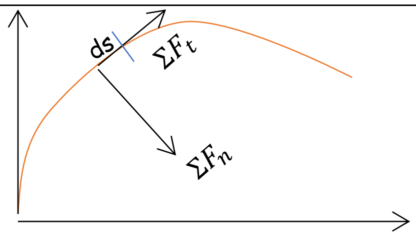
$$\frac{W}{g}v\, dv = \Sigma F_t.ds$$

Integrating both sides

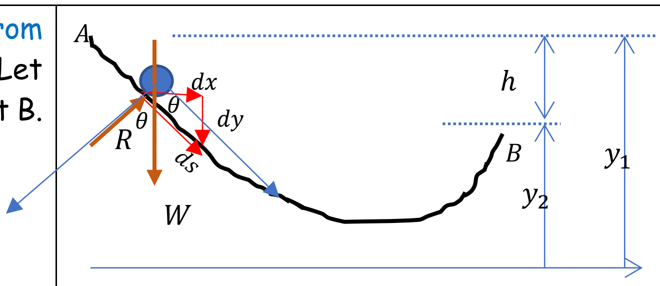
$$\frac{W}{g}\int_{v_0}^{v} v\, dv = \int_0^s \Sigma F_t\,.ds$$

$$\frac{W}{g}\frac{v^2}{2} - \frac{W}{g}\frac{{v_0}^2}{2} = \int_0^s \Sigma F_t\,.ds$$

Change in kinetic energy= net work done by tangential forces



Consider a body of weight W sliding down from rest along a curved plane without friction. Let it acquires a velocity v when it reaches point B.
The initial kinetic energy =0
Final kinetic energy= $\frac{W}{g}\frac{v^2}{2}$
The change in kinetic energy $= \frac{W}{g}\frac{v^2}{2} - 0$



The forces acting on the body are the weight W and reaction R. Let us assume that the body moves an infinitesimal distance ds along the plane. The normal reaction R and the component of weight $W\cos\theta$ are always perpendicular to the displacement ds and will not do any work over this displacement. The only force which is doing work is the component of weight $W\sin\theta$ which is in the same direction of ds.
Work done by the force $W\sin\theta$ over the small displacement ds,

$$= W\sin\theta \,.\, ds = W\left(-\frac{dy}{ds}\right)ds = -W.dy$$

Total work done by the forces when the body moves from A to B

$$= -\int_{y_1}^{y_2} W.dy = -W(y_2 - y_1) = Wh$$

By work-energy theorem, change in kinetic energy=net work done

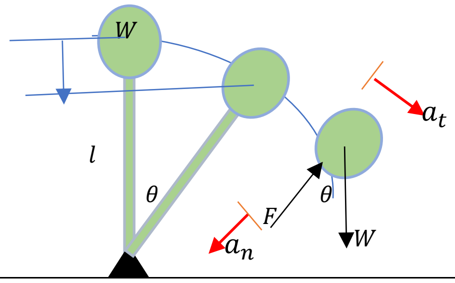
$$\frac{W}{g}\frac{v^2}{2} = Wh$$

$$v = \sqrt{2gh}$$

If friction is neglected, the speed that the body would gain in moving from A to B along any path is equal to the speed it would have gained had it fallen freely through a height h.

Example 1

A pendulum is released from rest in its position of unstable equilibrium as shown. Find the value of angle θ defining the position in its downward travel at which the axial force in the rod changes from compression to tension.



Let θ be the angle that the pendulum has swung down when the axial force in the bar changes from compression to tension. When a negative force changes to positive force, it has to go through a zero value. That is, at this inclination, the force in the bar is zero. If the angle is less than θ, the force is compressive and if it is greater than θ it is tensile.
The pendulum starts from rest.
Therefore, its initial kinetic energy is zero.
Let v be the velocity when it makes an angle θ.
Final kinetic energy $= \frac{W}{g}\frac{v^2}{2}$
Change in kinetic energy $= \frac{W}{g}\frac{v^2}{2} - 0$
Only two forces are acting on the system, the weight W and the axial force F in the bar. The axial force will be always normal to the circular path traced by the bob. Hence it will not do any work. The only force doing work is the weight W
Work done by the weight $= W \times (l - l\cos\theta) = Wl(1 - \cos\theta)$

By work-energy principle, change in kinetic energy= net work done

$$\frac{W}{g}\frac{v^2}{2} = Wl(1-\cos\theta) \ggg v = \sqrt{2gl(1-\cos\theta)}$$

If we can find out the velocity v, we can get the value of θ

We know that as the pendulum swings down, it will have both tangential and normal accelerations.

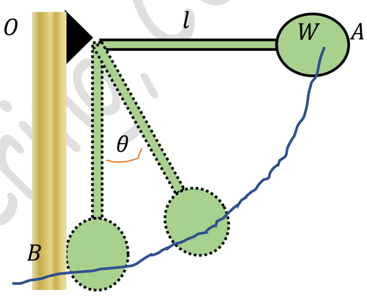
Writing down the equation of motion in the normal direction

$$\frac{W}{g}\frac{v^2}{l} = W\cos\theta - F \quad (But\ F = 0)$$

$$\frac{W}{g}\frac{v^2}{l} = W\cos\theta \quad \ggg \quad \frac{W}{g}\frac{2gl\,(1-\cos\theta)}{l} = W\cos\theta \ggg \ \theta = 48.2°$$

Example 2

A simple pendulum is released from rest in the horizontal position OA and falls in a vertical plane under the influence of gravity. If it strikes a vertical wall at B and the coefficient of restitution is $e = 0.5$, find the angle θ defining its total rebound.



The pendulum starts from rest. Therefore, its initial kinetic energy is zero.

Let v be the velocity when it strikes the wall.

Final kinetic energy $= \frac{W}{g}\frac{v^2}{2}$

Change in kinetic energy $= \frac{W}{g}\frac{v^2}{2} - 0$

Only two forces are acting on the system, the weight W and the axial force F in the bar. The axial force will be always normal to the circular path traced by the bob. Hence it will not do any work. The only force doing work is the weight W

Work done by the weight $= W \times l$

By work-energy principle, change in kinetic energy= net work done

$$\frac{W}{g}\frac{v^2}{2} = Wl \quad \ggg \quad v = \sqrt{2gl}$$

Let v' be the velocity of the bob after impact. Since the wall remains stationary $u = u' = 0$

$$v' - u' = -e(v-u) \quad \ggg \quad \therefore v' = -0.5\sqrt{2gl}$$

The velocity of the pendulum after impact is negative as it is rebounding

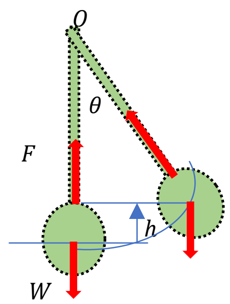
Now let us consider the motion of the pendulum after rebound

Initial kinetic energy of the pendulum$= \frac{W}{g}\frac{v'^2}{2} = \frac{W}{g}\frac{(0.5\sqrt{2gl})^2}{2} = 0.25Wl$

When it rebounds to maximum angle θ, the velocity of the pendulum become s zero.

$\therefore$ Final kinetic energy of the pendulum =0

Only two forces are acting on the system, the weight W and the axial force F in the bar. The axial force will be always normal to the circular path traced by the bob. Hence it will not do any work.



The only force doing work is the weight W

Work done by the weight $= -W \times h = -W(l - l\cos\theta) = -Wl(1-\cos\theta)$ (-ve because W & h are opposite in direction)

By work-energy principle, change in kinetic energy= net work done

$$0 - 0.25Wl = -Wl(1-\cos\theta) \quad \ggg \quad \cos\theta = 0.75 \quad \ggg\gg \quad \theta = 41.4°$$

Example 3

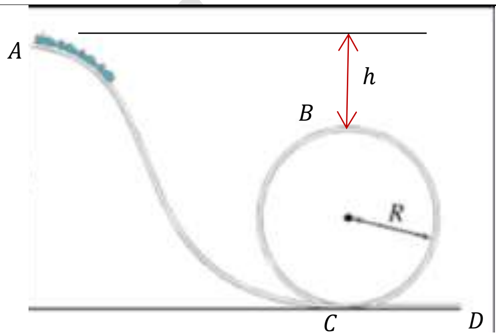
A small car of weight W starts from rest at A and rolls without friction along the loop-the-loop ACBD. What is the least height h above the top of the loop at which the car can start without falling off the track at B? For such a starting position, what velocity will the car have along the horizontal position CD of the track?

Let the car starts from A which is at a height h from the top of the loop B which is just sufficient to keep the car on the track at B.

Since friction is neglected, the velocity of the at B

$$v = \sqrt{2gh}$$

Let us consider the forces acting on the car at B. The forces are the weight W and reaction R. As the car is moving along a circular path, it will have a normal acceleration $a_n = \frac{v^2}{R}$ towards the centre of the circle. As no other force is there tangential to the path, its tangential acceleration is zero, and velocity is constant.



Writing down the equation of motion along the normal direction,

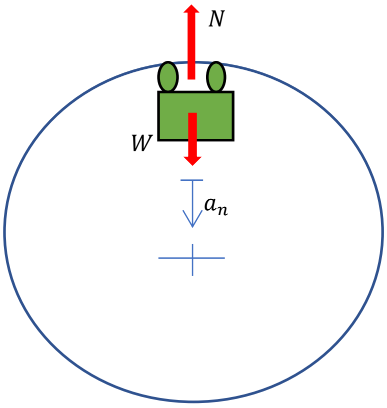
$$\frac{W}{g}\frac{v^2}{R} = W - N$$

But, the car starts from a height h such that its velocity is just sufficient to pass point B. That is, the car is just about to fall off the track at B. Then, the reaction R=0.

$$\frac{W}{g}\frac{(\sqrt{2gh})^2}{R} = W \quad \ggg \quad h = \frac{R}{2}$$

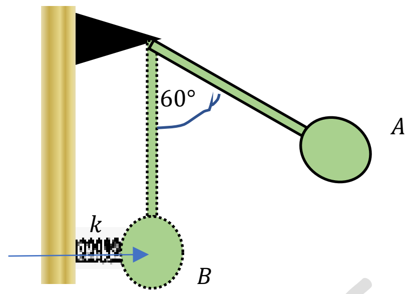
Hence, the car should start from a height $h > \frac{R}{2}$ in order not to fall off the track at B.

If the car starts at a height $h = \frac{R}{2}$, its velocity along the horizontal path=$\sqrt{2g(h+2R)} = \sqrt{5gR}$



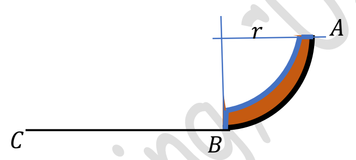
#Exercise 1

A simple pendulum of weight $W = 5\,N$ and length $l = 25\,cm$ is released from rest at A, swings downward under the influence of gravity and strikes a spring of stiffness $k = 1200\,N/m$ at B. Neglecting the mass of the spring, determine the compression that it will suffer.



#Exercise 2

A smooth circular tube AB in the form of a quarter circle of mean diameter r is fixed in a vertical plane and contains a flexible chain of length $\frac{\pi r}{2}$ and weight $\frac{\pi r w}{2}$. If released from rest in the configuration shown, find the velocity with which the chain will move along the smooth horizontal surface BC after it emerges from the tube.



#Exercise 3

In the figure, the pendulum A is released from rest in the horizontal position O_1A, swings down and strikes the pendulum B initially at rest in the vertical position. If the impact is perfectly elastic, find the value of angle θ when the axial force in the bar O_2B changes from compression to tension.

Find the angle of rebound φ of the pendulum A after impact.

Also determine the compression of the spring if it has a stiffness of 1200 N/m and negligible mass.

Take $W_a = 5\,N$, $W_b = 10\,N, l = 30\,cm$